Calc Proofs

Nathan Yan

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Abstract

Yeah, this is where I'll put my proofs in the future for safekeeping so I can reference them later or smth.

1 Limits

Limit proofs hinge on the following definition of a limit:

$$\lim_{x\to a} f(a) = L \iff \forall \varepsilon > 0,$$
$$\exists\, \delta \ s.t.\ 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon$$

1.1 Addition Rule

Statement:
given that

$$\lim_{x\to a} f(x) = L, \lim_{x\to a} f(x) = M$$

, show that

$$\lim_{x\to a} f(x) + g(x) = L + M$$

Proof. Consider some $\varepsilon > 0$, then by the definition of the limit there exists δ_1 and δ_2 such that:

$$0 < |x-a| < \delta_1 \implies |f(x) - L| < \frac{\varepsilon}{2}$$

$$0 < |x-a| < \delta_2 \implies |g(x) - M| < \frac{\varepsilon}{2}$$

Then let $\delta = \min(\delta_1, \delta_2)$, and we need

$$|f(x) + g(x) - (L + M)| < \varepsilon$$

We have:

$$0 < |x-a| < \delta$$

which implies

$$|f(x) - L| < \frac{\varepsilon}{2}$$

and

$$|g(x) - M| < \frac{\varepsilon}{2}$$

By the triangle inequality,

$$|a+b| \leq |a| + |b|$$

so

$$|f(x) + g(x) - (L+M)| < |f(x) - L| + |g(x) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

so

$$|f(x) + g(x) - (L+M)| < \varepsilon$$

and so limits follow the addition rule. ■ ■

Do note that I *did* need to consult the textbook for the triangle inequality stuff, but the rest was fine.

1.2 Multiplication Rule

Given that

$$\lim_{x\to a} f(x) = L$$

and

$$\lim_{x\to a} g(x) = M$$

prove that

$$\lim_{x\to a} f(x)g(x) = LM$$

Proof. ■

1.3 Polynomials are continuous

A function, $f(x)$, is continuous if

$$\lim_{x \to a} f(x) = f(a)$$

From this, let us prove that all polynomials of the form x^n for $n \in \mathbb{N}$ are continuous.

1.4 Rational Powers

Now, since a certain someone asked for it, I'll do *one* more epsilon delta proof.
Prove that

$$f(x) = \sqrt[n]{x}$$

is continuous for all $x > 0$

Proof. Consider some $\varepsilon > 0$, we have to find some δ such that:

$$0 < |x - a| < \delta \implies |\sqrt[n]{x} - \sqrt[n]{a}| < \varepsilon$$

for all $a > 0$.
Which is equivalent to

$$-\delta < x - a < \delta \implies -\varepsilon < \sqrt[n]{x} - \sqrt[n]{a} < \varepsilon$$

Technically $x \neq a$, but we don't need this condition.
Presume $\varepsilon < \sqrt[n]{a}$. This can always be satisfied because $a > 0 \implies \sqrt[n]{a} > 0$.
This implies $0 < \sqrt[n]{a} - \varepsilon < \sqrt[n]{a}$, and we can set $\delta = \min(a - (\sqrt[n]{a} - \varepsilon)^n, (\sqrt[n]{a} + \varepsilon)^n - a)$

$$-(a - (\sqrt[n]{a} - \varepsilon)^n) < x - a < (\sqrt[n]{a} + \varepsilon)^n - a$$
$$(\sqrt[n]{a} - \varepsilon)^n - a < x - a < (\sqrt[n]{a} + \varepsilon)^n - a$$
$$(\sqrt[n]{a} - \varepsilon)^n < x < (\sqrt[n]{a} + \varepsilon)^n$$

Yay! $\sqrt[n]{x}$ is a strictly increasing function, so we can safely:

$$\begin{align}
\sqrt[n]{x} - \sqrt[n]{a} &< \sqrt[n]{(\sqrt[n]{a} + \varepsilon)^n} + \sqrt[n]{a} \tag{1}\\
&= \sqrt[n]{a} + \varepsilon - \sqrt[n]{a} \tag{2}\\
&= \varepsilon \tag{3}
\end{align}$$

and then we also have

$$\sqrt[n]{x} - \sqrt[n]{a} > \sqrt[n]{(\sqrt[n]{a} - \varepsilon)^n} - \sqrt[n]{a} \qquad = \sqrt[n]{a} - \varepsilon - \sqrt[n]{a} = -\varepsilon \tag{4}$$

Which therefore implies that

$$-\varepsilon < \sqrt[n]{x} - \sqrt[n]{a} < \varepsilon$$

■

There is some weird domain stuffs, but I think I handled it fine. Figuring out the delta, I obviously needed the textbook for (my brain wasn't big enough), but for the rest I could figure it out...

2 Basic Derivatives

I hate epsilon delta proofs, no more! Derivative proofs are so much more intuitive, its great...

2.1 Power Rule for $n \in \mathbb{N}$

Easy proof with binomial theorem.

Proof. consider $f(x) = x^n$, $n \in \mathbb{N}$.

$$
\begin{aligned}
\frac{d}{dx}x^n &= \lim_{h\to 0} \frac{(x+h)^n - x^n}{h} && (5)\\
&= \lim_{h\to 0} \frac{-x^n + \binom{0}{n}x^n + \binom{1}{n}hx^{n-1} + \binom{2}{n}h^2n^{n-2} + \ldots + \binom{n-2}{h}^{n-2}n^2 + \binom{n-1}{n}h^{n-1}n + \binom{n}{n}h^n}{h} && (6)\\
&= \lim_{h\to 0} \frac{-x^n + x^n + nhx^{n-1} + \frac{n(n-1)}{2}h^2n^{n-2} + \ldots + \frac{n(n-1)}{2}h^{n-2}n^2 + nh^{n-1}n + h^n}{h} && (7)\\
&= \lim_{h\to 0} \frac{nhx^{n-1} + \frac{n(n-1)}{2}h^2n^{n-2} + \ldots + \frac{n(n-1)}{2}h^{n-2}n^2 + nh^{n-1}n + h^n}{h} && (8)\\
&= \lim_{h\to 0} \frac{h(nx^{n-1} + \frac{n(n-1)}{2}hn^{n-2} + \ldots + \frac{n(n-1)}{2}h^{n-3}n^2 + nh^{n-2}n + h^{n-1}}{h} && (9)\\
&= \lim_{h\to 0} nx^{n-1} + \frac{n(n-1)}{2}hn^{n-2} + \ldots + \frac{n(n-1)}{2}h^{n-3}n^2 + nh^{n-2}n + h^{n-1} && (10)\\
&= nx^{n-1} && (11)
\end{aligned}
$$

∎

Light work, no help needed.

2.2 Basic Trig

Here is $\cos x$:

Proof. Remember that:

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

Now consider our def of limit:

$$
\begin{aligned}
&\lim_{h\to 0} \frac{\cos(x+h) - \cos(x)}{h} && (12)\\
&= \lim_{h\to 0} \frac{\cos x\cos h - \sin x\sin h - \cos x}{h} && (13)\\
&= \lim_{h\to 0} \frac{\cos x(\cos h - 1) - \sin x\sin h}{h} && (14)\\
&= \lim_{h\to 0} \frac{\cos x(\cos h - 1)}{h} - \lim_{h\to 0} \frac{\sin x\sin h}{h} && (15)
\end{aligned}
$$

Now,

$$\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$$

and

$$\lim_{h\to 0} \frac{\sin h}{h} = 1$$

so it simplifies down to
$\cos x \cdot 0 - \sin x \cdot 1 = \boxed{\text{-sin x}}$ ∎

2.3 Quotient Rule

Quotient rule is pretty easy too. No help needed.

Proof.

$$
\begin{aligned}
\frac{d}{dx}\frac{f(x)}{g(x)} &= \lim_{h\to 0}\frac{\frac{f(x+h)}{g(x+h}-\frac{f(x)}{g(x)}}{h} && (16)\\
&= \lim_{h\to 0}\frac{f(x+h)g(x)-f(x)g(x+h)}{h\cdot g(x)g(x+h)} && (17)\\
&= \lim_{h\to 0}\frac{1}{g(x)g(x+h)}\cdot\frac{f(x+h)g(x)-f(x)g(x)+f(x)g(x)-f(x)g(x+h)}{h} && (18)\\
&= \lim_{h\to 0}\frac{1}{g(x)g(x+h)}\cdot\frac{g(x)(f(x+h)-f(x))+f(x)(g(x)-g(x+h))}{h} && (19)\\
&= \lim_{h\to 0}\frac{1}{g(x)g(x+h)}\cdot(g(x)\frac{f(x+h)-f(x)}{h}-f(x)\frac{g(x+h)-g(x)}{h}) && (20)\\
&= \lim_{h\to 0}\frac{1}{g(x)g(x+h)}\cdot(g(x)f'(x)-f(x)g'(x)) && (21)\\
&= \frac{g(x)f'(x)-f(x)g'(x)}{g(x)^2} && (22)
\end{aligned}
$$

■

2.4 Power Rule for $n \in \mathbb{Q}$

This one is solved in the textbook with implicit differentiation, but I provide my own proof here.

Proof. Recall that

$$a^n - b^n = (a-b)(a^{n-1} + ba^{n-2} + b^2a^{n-3} + \ldots + b^{n-2}a + b^{n-1})$$

The proof is trivial by "just distribute it"

Consider

$$x^{\frac{1}{n}}, n \in \mathbb{N}$$

$$
\begin{aligned}
\frac{d}{dx}x^{\frac{1}{n}} &= \lim_{h\to 0}\frac{(x+h)^{\frac{1}{n}}-x^{\frac{1}{n}}}{h} && (23)\\
&= \lim_{h\to 0}\frac{(x+h)^{\frac{1}{n}}-x^{\frac{1}{n}}}{h}\cdot\frac{(x+h)^{\frac{n-1}{n}}+(x+h)^{\frac{n-2}{n}}x^{\frac{1}{n}}+\ldots+(x+h)^{\frac{1}{n}}x^{\frac{n-2}{n}}+x^{\frac{n-1}{n}}}{(x+h)^{\frac{n-1}{n}}+(x+h)^{\frac{n-2}{n}}x^{\frac{1}{n}}+\ldots+(x+h)^{\frac{1}{n}}x^{\frac{n-2}{n}}+x^{\frac{n-1}{n}}} && (24)\\
&= \lim_{h\to 0}\frac{x+h-x}{h((x+h)^{\frac{n-1}{n}}+(x+h)^{\frac{n-2}{n}}x^{\frac{1}{n}}+\ldots+(x+h)^{\frac{1}{n}}x^{\frac{n-2}{n}}+x^{\frac{n-1}{n}})} && (25)\\
&= \lim_{h\to 0}\frac{h}{h((x+h)^{\frac{n-1}{n}}+(x+h)^{\frac{n-2}{n}}x^{\frac{1}{n}}+\ldots+(x+h)^{\frac{1}{n}}x^{\frac{n-2}{n}}+x^{\frac{n-1}{n}})} && (26)\\
&= \lim_{h\to 0}\frac{1}{((x+h)^{\frac{n-1}{n}}+(x+h)^{\frac{n-2}{n}}x^{\frac{1}{n}}+\ldots+(x+h)^{\frac{1}{n}}x^{\frac{n-2}{n}}+x^{\frac{n-1}{n}})} && (27)
\end{aligned}
$$

Now we can plug in 0 to get

$$
\begin{aligned}
&\frac{1}{nx^{\frac{n-1}{n}}} && (28)\\
&= \frac{1}{n}x^{\frac{1}{n}-1} && (29)
\end{aligned}
$$

Thus, the power rule holds for all nth roots.

Next, consider a function of the form

$$x^{\frac{a}{b}} (a, b \in \mathbb{N})$$

$$x^{\frac{a}{b}} \tag{30}$$
$$= (x^a)^{\frac{1}{b}} \tag{31}$$

Applying chain rule

$$\frac{d}{dx}((x^a)^{\frac{1}{b}}) = \frac{1}{b}(x^a)^{\frac{1}{b}-1}ax^{a-1} \tag{32}$$
$$= \frac{a}{b}x^{\frac{a}{b}-a}x^{a-1} \tag{33}$$
$$= \frac{a}{b}x^{\frac{a}{b}-a+a-1} \tag{34}$$
$$\frac{d}{dx}x^{\frac{a}{b}} = \frac{a}{b}x^{\frac{a}{b}-1} \tag{35}$$

Thus, the power rule holds for all rational powers. ■

Obviously, if I didn't use the textbook proof, I didn't need any help from it.

Regardless, I am going to provide the implicit differentiation proof here for reference (didnt need help from it though):

Proof. Consider $y = x^{\frac{a}{b}}$, where $a, b \in \mathbb{N}$

$$y^b = (x^{\frac{a}{b}})^b \tag{36}$$
$$y^b = x^a \tag{37}$$
$$by^{b-1}\frac{dy}{dx} = ax^{a-1} \tag{38}$$
$$\frac{dy}{dx} = \frac{ax^{a-1}}{by^{b-1}} \tag{39}$$
$$= \frac{ax^{a-1}}{b(x^{\frac{a}{b}})^{b-1}} \tag{40}$$
$$= \frac{ax^{a-1}}{bx^{a-\frac{a}{b}}} \tag{41}$$
$$= \frac{a}{b}x^{a-1-a+\frac{a}{b}} \tag{42}$$
$$= \frac{a}{b}x^{\frac{a}{b}-1} \tag{43}$$

■

Pretty simple, which is why the textbook used it.

3 Transcendentals

Chapter 3 was all applications, and chapter 4 was integrals and the proofs for FTC and whatnot all suck balls and require a lot of impossible yap, so we at chapter 5 now.

3.1 The Natural Log

The textbook first defines the natural log as is a function such that

$$\int_1^x \frac{1}{x} = \ln(x)$$

So we'll roll with this one. First,

$$\ln(1) = 0$$

Obviously.
Antiderivatives differ by at most a constant. Consider

$$\frac{d}{dx}(\ln(ax)) = \frac{a}{ax} = \frac{1}{x}$$

and

$$\frac{d}{dx}(\ln(x) + \ln(a)) = \frac{1}{x}$$

Therefore, these 2 expressions must differ by at most a constant. When $x = 1$, these functions both equal $\ln(a)$, so therefore they are equivalent and $\boxed{\ln(ab) = \ln(a) + \ln(b)}$

$$\ln(x^n) = \ln(\underbrace{x \cdot x \cdot x \cdot \ldots \cdot x}_{n}) = \boxed{n \ln x}$$

$$\ln(\frac{a}{b}) = \ln(ab^{-1}) = \boxed{\ln(a) - \ln(b)}$$

Now, it is time to combine this definition with the one learned in an algebra class, that is, $\ln x = \log_e(x)$. The proofs of logarithmic properties given a definition as the inverse of an exponential are trivial and will not be proven.

We need to show that there is some base, b, such that

$$\lim_{h\to 0} \frac{\log_b(x+h) - \log_b(x)}{h} = \frac{1}{x}$$

For all $x > 0$.

$$\lim_{h\to 0} \frac{\log_b(x+h) - \log_b(x)}{h} = \frac{1}{x} \tag{44}$$

$$\lim_{h\to 0} \frac{\log_b(\frac{x+h}{x})}{h} = \frac{1}{x} \tag{45}$$

$$\lim_{h\to 0} \frac{x}{h} \log_b(1 + \frac{h}{x}) = 1 \tag{46}$$

$$\lim_{h\to 0} \log_b((1 + \frac{h}{x})^{\frac{x}{h}}) = 1 \tag{47}$$

$$\tag{48}$$

Now we put the limit inside the log. This might seem objectionable, but log is continuous so its fine.

$$\log_b(\lim_{h\to 0}(1 + \frac{h}{x})^{\frac{x}{h}}) = 1 \tag{49}$$

$$\tag{50}$$

Let $u = \frac{x}{h}$. As $h \to 0^+$, $u \to \infty$, and when $h \to 0^-$, $u \to -\infty$, so we need to handle these separately and show that they converge to the same value.

$$\log_b(\lim_{u\to\infty}(1+\frac{1}{u})^u) = 1 \tag{51}$$
$$\tag{52}$$

Recall the compound interest definition of e:

$$\lim_{t\to\infty}(1+\frac{1}{t})^t$$

and we can apply it here.

$$\begin{aligned}
\log_b(e) &= 1 && (53)\\
e &= b^1 && (54)\\
b &= e && (55)
\end{aligned}$$

And so we are done with the positive case.

$$\log_b(\lim_{u\to-\infty}(1+\frac{1}{u})^u) = 1 \tag{56}$$
$$\tag{57}$$

Let $t = -u$. t approaches infinity now.

$$\begin{aligned}
\log_b(\lim_{t\to\infty}(1+\frac{1}{-t})^{-t}) &= 1 && (58)\\
\log_b(\lim_{t\to\infty}(\frac{t-1}{t})^{-t}) &= 1 && (59)\\
\log_b(\lim_{t\to\infty}(\frac{t}{t-1})^{t}) &= 1 && (60)\\
\log_b(\lim_{t\to\infty}(\frac{t-1+1}{t-1})^{t}) &= 1 && (61)\\
\log_b(\lim_{t\to\infty}(1+\frac{1}{t-1})^{t}) &= 1 && (62)\\
\log_b(\lim_{t\to\infty}(1+\frac{1}{t-1})^{t-1+1}) &= 1 && (63)\\
\log_b(\lim_{t\to\infty}(1+\frac{1}{t-1})^{t-1}\cdot(1+\frac{1}{t-1})) &= 1 && (64)\\
\log_b(\lim_{t\to\infty}(1+\frac{1}{t-1})^{t-1}\cdot\lim_{t\to\infty}(1+\frac{1}{t-1})) &= 1 && (65)\\
\log_b(e\cdot 1) &= 1 && (66)\\
\log_b(e) &= 1 && (67)\\
b &= e && (68)
\end{aligned}$$

Hooray! It works both ways, so we have successfully proven that $\boxed{\ln x = \log_e(x)}$

3.2 Exponentials

The book defines e^x as the inverse function of $\ln x$, so lets prove the derivative from that:

$$\ln y = x \tag{69}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 1 \tag{70}$$

$$\frac{dy}{dx} = y \tag{71}$$

$$\frac{dy}{dx} = e^x \tag{72}$$

Nice. From here, we can prove the general exponential:

$$y = a^x \tag{73}$$

$$\ln y = x \ln a \tag{74}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln a \tag{75}$$

$$\frac{dy}{dx} = \ln a \cdot a^x \tag{76}$$

In fact, this proof is so easy that is one of the few that we did in class.

4 Real analysis corner

So apparently I'm good for real analysis because I keep doing these proofs and they are always so easy.

4.1

Suppose that (p_n) and (q_n) are sequences in a metric space (X, d) with the property that $\lim_{n\to\infty} d(p_n, q_n) = 0$.

(i) Prove that if p_n is Cauchy then q_n is Cauchy.

Consider some $\varepsilon > 0$. We will now prove there exists some N such that for all $n, m > N$, $d(q_n, q_m) < \varepsilon$.

Because p_n is Cauchy, there exists some N_1 such that for all $n, m > N_1$, $d(p_n, p_m) < \frac{\varepsilon}{4}$.

Because $\lim_{n\to\infty} d(p_n, q_n) = 0$, there exists some N_2 such that for all $n, m > N_2$, $d(p_n, q_n) < \frac{\varepsilon}{4}$ and $d(p_m, q_m) < \frac{\varepsilon}{4}$

Then by triangle inequality,

$$d(p_m, q_n) < d(p_n, p_m) + d(p_n, q_n) < \frac{\varepsilon}{2}$$

and

$$d(p_m, q_m) < d(p_n, p_m) + d(p_m, q_m) < \frac{\varepsilon}{2}$$

And then we can combine it to get

$$d(q_n, q_m) < d(p_m, q_n) + d(p_m, q_m) < \varepsilon$$

And we are done!

4.2

Prove that for an decreasing sequence a_n the infinite sum of a_n converges $\iff$ $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Firstly, a_n must be non negative. Suppose that some a_N is negative. Then, for all $n > N$, a_n is negative. This sum clearly diverges to $-\infty$.

Then, the sum of the sequence must converge to a nonnegative number.

Let us consider

$$\frac{\sum_{k=0}^{\infty} 2^k a_{2^k}}{2} = \sum_{k=0}^{\infty} 2^{k-1} a_{2^k}$$

If $\sum_{k=0}^{\infty} 2^{k-1} a_{2^k}$ converges, then $\sum_{k=1}^{\infty} 2^{k-1} a_{2^k}$ converges because it is off by a constant.

Let's expand this expression and break up all the 2^k terms, to get

$$a_2, a_4, a_4, a_8, a_8, a_8, a_8, \ldots a_{2^{\lceil \log_2(i+1) \rceil}} \ldots$$

$$a_1, a_2, a_3, a_4, a_5, a_6, a_7, \ldots a_i \ldots$$

It is clear that the $i \leq 2^{\lceil \log_2(i+1) \rceil}$, and since this is a decreasing sequence it implies that

$$0 < \sum_{k=1}^{\infty} 2^{k-1} a_{2^k} < \sum_{i=1}^{\infty} a_i$$

Yay it converges! Which, of course also implies that $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

You can use the same process to prove the other side of the biconditional by proving that

$$0 < \sum_{i=1}^{\infty} a_i < \sum_{k=0}^{\infty} 2^k a_{2^k}$$